Mubashir Ali

Researcher — Bio Medical Scientist — AI Innovator

Islamabad, Pakistan | +92 343 5450454
mubashirali1837@gmail.com | mubashirali.bio
github.com/mubashir1837 | linkedin.com/in/mubashirali3

Professional Summary

Aspiring Computer Science and Bioinformatics Researcher, Google Knowledge Panel Holder, with a strong focus on Artificial Intelligence, Machine Learning, and CRISPR-Cas9 gene-editing technologies. I build computational tools that enhance the precision, speed, and efficiency of modern genomics workflows. My work includes developing AI-powered platforms for optimized guide RNA design, accurate off-target prediction, and faster therapeutic discovery. With hands-on experience at the intersection of biology and computation, I aim to push forward next-generation genome engineering solutions that support real-world medical and research applications.

Education

University of the People, California **2024 – 2028**
Bachelor of Science Computer Science
GPA 3.8

Uswa College, Islamabad **2021 – 2023**
Intermediate, Medical
Graduated with distinction (Top 5%)

Uswa Public School and College, Skardu **2019 – 2021**
Secondary School Education, Science

Experience

International Model United Nations **Sep 2025 – Present**

- Represented International Model United Nations (IMUN) on campus by promoting global diplomacy, leadership, and youth engagement initiatives.
- Coordinated outreach campaigns to increase student participation in IMUN conferences, webinars, and international programs.
- Facilitated communication between IMUN organizers and university students, ensuring timely dissemination of opportunities and events.
- Supported event planning and digital promotion through social media and peer networks, contributing to increased awareness and registrations.

Code with Bismillah – Founder **Jun 2023 – Present**
Islamabad, Pakistan

TynexAI – Founder **April 2025 – Present**
Islamabad, Pakistan

- Founded a tech Education startup developing innovative software solutions and give tech education.
- Lead a team of 3 developers creating custom web applications for research institutions
- Implemented agile methodologies resulting in 40% faster project delivery times

Aspire Leaders Program – Participant **December 2025 – Present**

- Selected for a prestigious leadership development program focused on strategic thinking, community impact, and personal growth.
- Engaged in workshops and projects aimed at enhancing problem-solving, teamwork, and leadership skills.
- Collaborated with peers on social initiatives and innovation challenges, applying real-world solutions to local and global issues.

- Presented project outcomes to mentors and program coordinators, receiving feedback to improve execution and communication.

Innvo AI – Generative AI Content Creator **Oct 2024 – Present**
Islamabad, Pakistan

- Produce technical content explaining complex AI concepts

YouTube – Content Creator **Jun 2023 – Present**

- Created 100+ educational videos on technology, and AI/ML applications
- Developed interactive tutorials demonstrating practical applications of technology

MLSA Youth Empowerment – Technical Host & Workshop Facilitator **Feb 2023 – Present**
Islamabad, Pakistan

- Organized 15+ workshop
- Mentored 50+ students in developing technical skills

Behance & Fiverr – Freelance Software Developer **Sep 2023 – Present**

- Specialized in bioinformatics tools, AI integrations, and genomic data visualization dashboards

Featured Projects

- **Genonme House**
  Modular bioinformatics toolkit for sequence analysis, parsing, ML, and visualization
- **GeneFix AI**
  Machine learning model predicting CRISPR-Cas9 editing efficiency and potential off-target effects to design more precise gene editing experiments.
- **Bio Data Hub (VS Code Extension)**
  Visual Studio Code extension designed for bioinformatics professionals and data scientists. It simplifies the exploration, visualization, and management of CSV datasets, enabling users to analyze biological data efficiently.
- **Biobuntu - Ubuntu base Bioinformatics Pipeline**
  A comprehensive platform for running genomic analysis pipelines with CLI, GUI, and web interfaces. Perfect for researchers, labs, and bioinformatics teams.
- **Protein Structure Prediction**
  Computational pipeline using deep learning to predict protein structures, analyze amino acid sequences, and identify potential binding sites.
- **Heart Disease Prediction ML**
  Developed ML model on Azure predicting heart disease risk with 89% accuracy. Created user-friendly interface for healthcare professionals.
- **BismillahCSS (NPM Package)**
  Utility-first CSS framework optimized for futuristic web applications.
- **PK Weather (Gemini AI Integration)**
  Advanced weather application powered by Google's Gemini AI API providing intelligent forecasting with natural language processing capabilities.

Technical Skills

Programming Languages

Python, JavaScript, TypeScript, R, SQL, HTML/CSS, C/C++, Java, Assembly, LaTeX

DeveOps tools

Azure Machine Learning Deployment, Azure Container, GitHub Actions, NPM, VS Code Publisher, Docker, Vercel, Netlify, Heroku

Frameworks & Libraries

React.js, Next.js, Node.js, TensorFlow, PyTorch, Pandas, BioPython, Scikit-learn, NumPy, GraphQL, Spark

Bioinformatics Tools

BLAST, Clustal, Ensembl, UCSC Genome Browser, Galaxy, CRISPR Design Tools

Development Tools

Git, GitHub, VS Code, WSL, Docker, Jupyter, AWS, Azure, Google Colab, Power BI

Microsoft Office

Power BI (data handling (Dataset, Data Model), transformation (Power Query, ETL), analysis (DAX, Measure, Calculated Column), visualization (Report, Dashboard, Visual, Tile, Filter,

Slicer), and sharing (Service, Gateway, Publish) Ms Excel, Word

Database Technologies

MongoDB, PostgreSQL, Firebase, MySQL, SQLServer, GraphQL

Achievements & Research Interests

Certifications

- Data Science Specialization (Coursera IBM)
- Google Advanced Data Analytics Specialization (Coursera)
- Genetic Data Science Specialization (Johns Hopkins University - Coursera)
- AI in Health Care by Stanford University
- Generative AI (Google Cloud)
- Bioinformatics - Coursera
- Biology Meets Programming: Bioinformatics
- Foundations of Coding Full-Stack (Microsoft)
- Front-End Developer Capstone (Meta)
- CS302:Software Engineering (Syalor)
- Coding Full-Stack (Microsoft Coursera)
- Artificial Intelligence (IBM)

Achievements

- Google Knowledge Panel Holder – Recognized as a verified profile on Google Knowledge Panels

Research Interests

- AI-driven optimization of CRISPR-Cas9 guide RNA design
- ML approaches for predicting gene editing outcomes
- Computational tools for reducing off-target effects
- Deep learning applications in genomic data analysis
- Predictive modeling for disease diagnosis

Publications & Languages

Publications

- GeneFix-AI: AI-Powered CRISPR-Cas9 System for Real-Time Detection and Correction of Mutations in Non-Human Species
- Direct Brain Knowledge Transfer Interface (DBKTI): A New Paradigm for Neural Knowledge Uploading
- BioDataHub: An Integrated VS Code Extension for Streamlined Bioinformatics Dataset Analysis and Visualization." *researchsquare*

Languages

- **Urdu:** Native
- **English:** Professional Working